

The Kepler–Bouwkamp partial products do not have cubic convergence

ColdSmile

Disproof of conjecture 00000000283

4 October 2026

Abstract

For the ordinary partial products $S_N = \prod_{k=3}^N \cos(\pi/k)$ and their strictly positive infinite-product limit K , we prove

$$S_N - K \geq \frac{2K}{(N+1)^2} \quad (N \geq 3).$$

Consequently, for every real constant C and every cutoff N_0 , some $N \geq \max(N_0, 3)$ satisfies $|S_N - K| > C/N^3$. In particular, the error is not $O(N^{-3})$. This refutes the convergence-rate conjunct of the stated conjecture. The argument does not decide the transcendence or Gamma-value claims and does not assert a sharp asymptotic rate.

1 The exact statement and the actual products

Both language versions of conjecture 00000000283 define the Kepler–Bouwkamp constant by the real cosine product

$$K_B = \prod_{k \geq 3} \cos\left(\frac{\pi}{k}\right)$$

and assert that its partial products converge at rate n^{-3} , alongside transcendence and a proposed Gamma-function reduction. We refute the rate assertion for these ordinary, uncorrected partial products. Its usual upper-bound interpretation requires constants $C \geq 0$ and $N_0 \in \mathbb{N}$ such that

$$|S_N - K_B| \leq \frac{C}{N^3} \quad \text{for every } N \geq \max(N_0, 3). \quad (1)$$

The source does not introduce an accelerated product or a subtracted asymptotic correction. A stronger interpretation such as an exact cubic order would also entail (1).

For a natural-number indexing starting at zero, put

$$a_j = \cos\left(\frac{\pi}{j+3}\right), \quad P_n = \prod_{j=0}^{n-1} a_j, \quad K = \prod_{j=0}^{\infty} a_j. \quad (2)$$

The last expression is the actual infinite product, whose convergence and positivity will be proved below. The conventions give $P_0 = 1$ and $P_1 = \cos(\pi/3)$. Define the source-indexed product using the integer interval,

$$S_N = \prod_{k \in \{3, \dots, N\}} \cos\left(\frac{\pi}{k}\right).$$

Empty intervals have product 1. The exact reindexing is

$$S_{n+2} = P_n \quad (n \in \mathbb{N}), \quad S_N = P_{N-2} \quad (N \geq 3). \quad (3)$$

Thus the proof concerns exactly the truncations appearing in the conjecture. The Lean project proves the first identity by induction, using the actual product over `Finset.Icc 3 N`; its error theorem then applies it with $N = n + 2$.

2 A positive, convergent infinite product

Lemma 1. *For every $j \in \mathbb{N}$, $1/2 \leq a_j \leq 1$ and*

$$\frac{2}{(j+3)^2} \leq 1 - a_j \leq \frac{\pi^2}{2(j+3)^2}. \quad (4)$$

Proof. Set $x = \pi/(j+3)$. Then $0 < x \leq \pi/3$, so monotonicity of cosine on $[0, \pi]$ gives $a_j \geq \cos(\pi/3) = 1/2$, and $a_j \leq 1$. For $y = x/2 \in [0, \pi/2]$, the sine bounds

$$\frac{2y}{\pi} \leq \sin y \leq y$$

follow respectively from concavity of sine and its tangent bound. Using $1 - \cos x = 2 \sin^2(x/2)$, we obtain

$$\frac{2x^2}{\pi^2} \leq 1 - \cos x \leq \frac{x^2}{2}.$$

Substitution of $x = \pi/(j+3)$ proves (4). □

The upper bound and the convergent reciprocal-square series give

$$\sum_{j=0}^{\infty} |a_j - 1| < \infty.$$

For completeness, logarithm summability follows quantitatively: if $1/2 \leq a \leq 1$, then

$$0 \leq -\log a = \int_a^1 \frac{dt}{t} \leq 2(1 - a).$$

Consequently,

$$\sum_{j=0}^{\infty} |\log a_j| \leq \sum_{j=0}^{\infty} \frac{\pi^2}{(j+3)^2} < \infty.$$

Write $L = \sum_{j=0}^{\infty} \log a_j$. Each factor is positive, so finite-product logarithms and continuity of the exponential imply

$$P_n = \exp \left(\sum_{j=0}^{n-1} \log a_j \right) \longrightarrow \exp(L) = K > 0. \quad (5)$$

Absolute summability also certifies the infinite product itself: exponentiating finite logarithm sums over arbitrary finite sets gives the same product limit. In Lean, `constant` is defined by `tprod`; the separate theorem `hasProd_factor` establishes convergence to it, and `constant_eq_exp_logSum` establishes (5). Thus a totalized product definition is not being used as a substitute for convergence. By (3), $S_N \rightarrow K$, and this K is the source's K_B .

3 The first omitted factor forces an error

The finite products satisfy

$$P_{n+1} = P_n a_n, \quad P_n > 0, \quad P_{n+1} \leq P_n.$$

Their limit therefore obeys $K \leq P_n$ for every n . In particular,

$$\begin{aligned} P_n - K &\geq P_n - P_{n+1} = P_n(1 - a_n) \\ &\geq K(1 - a_n) \geq \frac{2K}{(n+3)^2}. \end{aligned} \quad (6)$$

The second inequality uses both $P_n \geq K$ and $1 - a_n \geq 0$. For $N \geq 3$, substitute $n = N - 2$ in (6) and use (3) to obtain the claimed source-indexed bound:

$$0 < \frac{2K}{(N+1)^2} \leq S_N - K = |S_N - K| \quad (N \geq 3). \quad (7)$$

Only the first omitted factor was needed. This lower bound suffices for the disproof; it is not a matching upper bound or a claim that the true error has exact order N^{-2} .

4 Every proposed cubic bound fails

Theorem 1. *For every $C \in \mathbb{R}$ and every $N_0 \in \mathbb{N}$, there exists a natural number $N \geq \max(N_0, 3)$ such that*

$$\frac{C}{N^3} < |S_N - K|.$$

Equivalently, the sequence $N^3|S_N - K|$ is unbounded on every tail. In particular,

$$(S_N - K)_{N \in \mathbb{N}} \text{ is not } O(N^{-3}) \quad (N \rightarrow \infty).$$

Proof. For $N \geq 3$, we have $N + 1 \leq 2N$. Multiplying (7) by $N^3 > 0$ therefore yields

$$N^3|S_N - K| \geq \frac{2KN^3}{(N+1)^2} \geq \frac{K}{2}N. \quad (8)$$

Because $K > 0$, choose an integer $N \geq \max(N_0, 3)$ with $N > 2C/K$. Then (8) gives $N^3|S_N - K| > C$. Division by the positive number N^3 gives the desired violation. Since C and N_0 were arbitrary, no eventual cubic upper bound can hold. $\square$

This argument rules out the rate conjunct and hence the conjunction asserted by conjecture 00000000283. It leaves the transcendence of K_B and the suggested reduction to algebraic independence of Gamma values undecided. It uses neither numerical approximations to K_B nor a special-function representation of it.

5 Formalization and scope

The accompanying project uses Lean 4.19.0 and its pinned Mathlib dependencies. Its objects are real cosine factors, ordinary finite products, and the genuine infinite product. `Factors.lean` proves the factor estimates, logarithm summability, a `HasProd` certificate, positivity, and convergence of finite prefixes. `Prefixes.lean` proves monotonicity, the exact interval-product bridge, convergence of the source-indexed products, and (7).

In `Rate.lean`, `cubic_bound_counterexample` has the unrestricted quantifiers of Theorem 1. The definition `CubicConvergence` is Mathlib’s actual `Asymptotics.IsBigO` at `atTop`, applied to

$$N \mapsto S_N - K, \quad N \mapsto ((N : \mathbb{R})^3)^{-1}.$$

Its norm-based definition gives the ordinary eventual absolute-error inequality. The cutoff in the quantitative theorem is at least 3, so no division-by-zero convention at $N = 0$ enters the contradiction. The final theorem `conjecture_283` proves $\neg \text{CubicConvergence}$. It formalizes the false rate conjunct rather than introducing a substitute for the unspecified Gamma-value reduction.

Build, strict replay, axiom inspection, source-identity checks, and independent semantic review are documented separately in the submission’s verification materials. Those local records do not claim acceptance by the competition’s maintainers.

References

- [1] M. Chamberland and A. Straub, *On gamma quotients and infinite products*, arXiv:1309.3455 (2013), Section 4.1. arxiv.org/abs/1309.3455. This section provides background on the Kepler–Bouwkamp product and its numerical evaluation. No theorem or numerical result from it is assumed in the proof above.